

Technical Specification: AMACE Architectural Workflow and Component Logic

1. Architecture Core Concept: The Three-Stratum Model

The AMACE (Autonomous Multi-Stratum Anomaly Correlation Engine) is engineered to eliminate the "Cross-Stratum Blind Spot"—a systemic failure in traditional monitoring where tools operate within isolated layers, unable to correlate simultaneous deviations that share a common causal origin. AMACE formalizes the computing environment into a Three-Stratum Model, enabling domain-agnostic causal discovery.

Stratum	Formal Definition	Data Characteristics	Volume & Structure
\$S_0\$: Observation	The lowest abstraction layer containing raw, unprocessed event data.	Unstructured/semi-structured; no semantic interpretation.	High volume (10^6 – 10^{12} events/day).
\$S_1\$: Representation	Intermediate layer containing computed statistical aggregations and derived metrics.	Structured; represents engineered knowledge and feature distributions.	Moderate volume.
\$S_2\$: Inference	The highest abstraction layer containing outputs from statistical or ML models.	Highly structured; represents predictive intelligence and interpretations.	Low volume.

2. The AMACE Seven-Module Pipeline Overview

AMACE executes a strict linear pipeline to transform heterogeneous, domain-agnostic data into validated causal insights. By standardizing diverse telemetry into a universal tensor format, the system moves beyond simple alert clustering into formal causal proof.

Architectural Rationale: Complexity Reduction The primary technical breakthrough of AMACE v2 is the reduction of computational complexity for cross-stratum correlation. Traditional exhaustive search across N anomalies with D dimensions of cardinality K is $O(N^2 \times K^D)$, which is computationally intractable for production environments. AMACE reduces this to $O(N \log N \times k \log D)$, where k is the number of informationally relevant dimensional slices. This represents an exponential improvement, making real-time causal discovery feasible in high-cardinality systems.

3. Module 1: Stratum Schema Inference Engine (SSIE)

Input: Raw streaming data from domain-agnostic sources. **Transformation Logic:** SSIE utilizes **Entropy-Based Field Classification** to assign functional roles to data fields without manual configuration. It computes a 4-element probability vector for each field f using the following sigmoid functions:

Classification Role	Formal Probability Computation	Statistical Fingerprint
Entity Identifier	$P_{\{entity\}}(f) = \sigma(\alpha_1 H(f) + \alpha_2 \frac{1}{ unique(f) })$	Timestamp $P_{\{timestamp\}}(f) = \sigma(\beta_1 mon(f) + \beta_2 per(f) + \beta_3 parse(f))$ Monotonicity, periodicity, and datetime parseability.
Dimension	$P_{\{dimension\}}(f) = \sigma(\gamma_1 (1 - \frac{1}{ unique(f) }))$	Metric $P_{\{metric\}}(f) = \sigma(\delta_1 num(f) + \delta_2 cont(f) + \delta_3 var(f))$ Numerical continuity and high temporal variance.

Output: A vector $P_{\{entity\}}, P_{\{timestamp\}}, P_{\{dimension\}}, P_{\{metric\}}$ used for downstream structural mapping.

4. Module 2: Stratum-Adaptive Deviation Detector (SADD)

Input: Classified data fields from Module 1. **Transformation Logic:** SADD employs detection mechanisms specifically calibrated for the statistical profiles of each stratum:

- **S_0 (Observation):** Modified Z-score with **Median Absolute Deviation (MAD)** ($\text{threshold} = \text{median} + k \cdot \text{MAD}$) to ensure robustness against high-volume outliers.
 - **S_1 (Representation):** **Population Stability Index (PSI)** with two-sample **Kolmogorov-Smirnov confirmation** to detect distribution shifts.
 - **S_2 (Inference):** **Prediction Interval Monitoring** to identify model outputs falling outside calibrated interpretation bounds.
- Output: The Deviation Signature Tensor (DST)** The DST is a novel, universal algebraic structure defined as a 7-dimensional tuple: $\text{DST} = (S, E, T, D, M, C, \Phi)$.
- **S (Stratum):** Origin identifier $\{S_0, S_1, S_2\}$.
 - **E (Entity Set):** Canonical set of affected entity identifiers.
 - **T (Temporal Envelope):** Quad-timestamp signature (onset, peak, decay, duration).
 - **D (Dimensional Locus):** Dimension-value pairs defining the affected sub-population.
 - **M (Magnitude Vector):** Multi-scale measures (absolute, relative, z-score).
 - **C (Confidence):** Statistical bounds $[\mu - z\sigma, \mu + z\sigma]$.
 - **Φ (Provenance):** SHA-256 hash for cryptographic auditability.
- Architectural Rationale:** Unlike JSON-based signals, the DST allows for formal algebraic operations (union, intersection, and containment), enabling standardized comparison across disparate domains.

5. Module 3: Entropic Relevance Decomposition (ERD)

Input: DSTs from Module 2. **Transformation Logic:** ERD identifies the most relevant dimensional slice (σ) by maximizing the objective function: $\text{ERD}(\sigma) = \text{Surprise}(\sigma) - \text{Cost}(\sigma)$.

- **$\text{Surprise}(\sigma)$:** $-\log_2(m_{\text{slice}} / m_{\text{parent}}) \times |E_{\text{slice}}|$.
 - **$\text{Cost}(\sigma)$:** $\text{depth}(\sigma) \times \log_2(\text{cardinality}(\sigma))$.
- Process: Beam Search with Entropic Pruning**
1. **Initialize:** Create a beam B_0 of top- w single-dimension slices.
 2. **Expand:** Iteratively add dimensions to candidates.
 3. **Prune:** Discard candidates where $\text{Cost}(\sigma) > \text{Surprise}(\sigma)$. This "Entropic Pruning" ensures the system only retains slices that are "worth" their own description.
 4. **Terminate:** Return σ^* that maximizes the score within $O(w \times D \times K_{\text{max}})$ complexity.

6. Module 4: Cross-Stratum Causal Tensor (CSCT)

Input: DST pairs and relevant dimensional slices. **Transformation Logic:** The CSCT evaluates causality using a **Four-Axis Similarity Tensor**. | Axis | Symbol | Formal Computation | Novel Element || ----- | ----- | ----- | ----- || **Entity Coincidence** | α | $J(E_A, E_B) \times \lambda$ | **Surprise-adjusted Jaccard** : λ multiplies the score by $\log_2(\text{overlap}_{\text{obs}} / \text{overlap}_{\text{exp}})$. || **Temporal Precedence** | β | $G(\Delta t, \mu, \sigma)$ | **Learned Causal Delay** : Gaussian kernel G based on historically learned stratum propagation delays. || **Dimensional Coherence** | γ | $\sum w_d \times \text{match}(D_A, D_B)$ | **Hierarchical Matching** : Recognizes parent-child matches (e.g., "Region" to "City"). || **Semantic Affinity** | δ | $\cos(\text{embed}_A, \text{embed}_B)$ | **Self-trained Embeddings** : Learned via contrastive co-occurrence (e.g., error rate vs. latency). |

Normalization: The four axes are combined via **Spectral Normalization** : $\text{CSCT}(A,B) = \frac{\sum w_i \cdot \sigma_i(A,B) \cdot \hat{e}_i}{\|\sum w_i \cdot \sigma_i(A,B) \cdot \hat{e}_i\|}$. The system calculates the final score based on the **Riemannian manifold distance** , ensuring geometric integrity across the orthogonal axes.

7. Module 5: Computational Ablation Protocol (CAP)

Input: High-scoring CSCT correlations. **Transformation Logic:** To move from correlation to causation, AMACE performs three formal counterfactual tests:

1. **Exclusivity:** Remove DST_A ; if DST_B remains, the link is coincidental. *Criterion:* Anomaly reduction ratio > 0.3 .
2. **Sufficiency:** Use DST_A to predict DST_B magnitude. *Criterion:* Prediction error < 0.5 .
3. **Minimality:** Iteratively remove nodes in S groups to find the smallest set required for explanation. **Statistical Rigor:** Each test runs with **1000-iteration bootstrapped confidence intervals** . A relationship is only validated if the criterion holds at the **95% confidence level** ($\alpha = 0.05$).

8. Module 6: Causal Insight Graph Synthesizer (CIGS)

Input: Validated relationships from Module 5. **Transformation Logic:** CIGS assembles relationships into a Directed Acyclic Graph (DAG) representing the causal cascade across strata. **Output: Causal Insight Object Checklist**

- **Root Cause:** The initiating S_0 deviation.
- **Propagation Path:** The intermediate S_1 nodes.
- **Terminal Effects:** Observable S_2 impacts.
- **Entity Intersection:** Common entities affected across the chain.
- **Aggregate Confidence:** Composite score based on similarity and ablation success.

9. Module 7: Predictive Precursor Engine (PPE)

Input: Current DSTs and the Precursor Sequence Library (PSL). **Transformation Logic:** PPE matches current DST "prefixes" against historical causal graphs in the PSL. **Probability**

Formula: $P(\text{incident} | \text{prefix}) = \frac{\text{count}(\text{prefix} \rightarrow \text{incident})}{\text{count}(\text{prefix})} \times \text{temporal_decay}(\text{age})$

Anti-Correlation Detection: The PPE includes a critical "Anti-Correlation" module. It monitors for the *absence* of expected correlations (e.g., a latency

spike in S_1 failing to trigger an autoscaler event in S_2). Flagging these "missing" links allows AMACE to detect latent infrastructure bugs, such as misconfigured health checks or broken automation logic, before they manifest as outages.

10. Technical Complexity and Performance Summary

The AMACE v2 specification represents a transition from conceptual heuristics to a mathematically rigorous causal discovery framework.

Feature	v1	Cross-Layer System
AMACE v2 Specification	-----	-----
Data Structure	Ad-hoc JSON	Deviation Signature Tensor (DST) : 7-dim algebraic tuple
Search Complexity	$O(N^2 \times K^D)$ (Exponential)	$O(N \log N \times k \log D)$ (Log-linear)
Dimensional Pruning	Conceptual MDL	ERD with formal convergence proof
Correlation Logic	Weighted sum	Four-Axis Tensor with Riemannian normalization
Validation	Narrative/Manual	Computational Ablation Protocol (CAP)
Configuration	Manual Discovery	SSIE : Autonomous entropy-based inference
Anomalies	Positive correlations only	Anti-Correlation Detection (identifying missing links)